PulmoTwin: Outpatient Respiratory Digital Twin

Slide 1: Title Slide

A 48-Hour Predictive Early Warning System for COPD & Asthma Flare-Ups

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- Team Name: <MY TEAM NAME>

- GitHub: https://github.com/MonalikaDoda/PulmoTwin
- Competition: Happiest Health Digital Twin Challenge 2026
- Stack: Python (FastAPI, XGBoost, SHAP) + React Dashboard (Vite, Three.js)
- License: MIT License | Research Prototype

The Crisis of Reactive Respiratory Care in Delhi NCR

Slide 2: Clinical Problem Statement

- Clinical Reality: Chronic respiratory diseases are a leading cause of emergency hospitalization in India. (Reference: Lancet Global Health / ICMR State-Level Disease Burden Initiative, 2018).
- Compound Urban Stressors: Delhi NCR experiences severe seasonal air quality episodes (CPCB NAQI > 300), extreme diurnal temperature swings, and mineral dust.
- The Critical Window: Airway inflammation begins 48–72 hours prior to acute decompensation (gradual SpO2 drift, mild tachycardia, nocturnal sleep reduction, and step count drop).
- The Gap: Current care is reactive—treating patients only after severe dyspnea forces an ER visit.

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PulmoTwin: 48-Hour Predictive Outpatient Digital Twin

Slide 3: Proposed Solution

- Multi-Modal Fusion: Fuses personal baseline vitals (SpO2, HR, RR, sleep, steps) with real modelled air quality (Open-Meteo CAMS) and weather (Open-Meteo ERA5).
- 48-Hour Early Warning: Identifies pre-flare physiological drift before acute hospitalization is required.
- Operational Feasibility: 0.77 alert days per patient per month (~9.19 alert days/patient/year), grouping consecutive days into multi-day episodes to prevent alert overload.
- Transparent Attribution: SHAP factor decomposition shows what shifted the estimate from the baseline.
- Counterfactual Sensitivity: Interactive What-If sliders show model response under hypothetical inputs.

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Real Modelled Weather/AQI + Synthetic Clinical Cohort

Slide 4: Data Sources & Provenance

- Synthetic Cohort (N=500 patients): Uses GOLD/GINA stage categories; proportions are our assumptions.
- Synthetic Telemetry (182,500 patient-days): 365 daily records with simulated pre-flare drift and noise.
- Real Modelled Air Quality: Open-Meteo CAMS reanalysis mapped to Indian CPCB NAQI breakpoints (CC BY 4.0).
- Real Modelled Weather: Open-Meteo ECMWF ERA5-Land reanalysis (temperature, diurnal swing, dust, rain).
- Illustrative Pollen Index: Mathematical curve placed near seasons in Delhi pollen calendars; not measured data.
- Ethical Standard: Zero real patient data (no PHI). All patient data is synthetic.

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Empirical Benchmarks from model/comprehensive_metrics.json

Slide 5: Multi-Modal Ablation Study

Strict patient-level GroupShuffleSplit (400 train / 100 test patients; 36,500 test days; 0 leakage):

- M1 (Vitals Only, 17 feat): ROC-AUC 0.9982 | PR-AUC 0.9042 | Recall 94.09% | F1 0.8353
- M2 (Vitals + EHR, 27 feat): ROC-AUC 0.9981 | PR-AUC 0.9020 | Recall 93.82% | F1 0.8400
- M3 (Vitals + EHR + CAMS AQI, 35 feat): ROC-AUC 0.9982 | PR-AUC 0.9068 | Recall 93.82% | F1 0.8374
- M4 (Full Multi-Modal, 46 feat): ROC-AUC 0.9982 | PR-AUC 0.9078 | Recall 93.41% | F1 0.8358

Plain Finding: On this synthetic dataset, continuous vitals (M1) already yield PR-AUC 0.9042.
Adding environmental features (M4) provides virtually negligible gain (+0.0036 PR-AUC, +0.0005 F1),
showing that vital sign drift accounts for almost all predictive signal in the generator.

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Performance Under Warning Horizons and Missing Data

Slide 6: Event-Level Advance Detection vs Day-Level Recall

Lead-Time Analysis (372 test flare events in test cohort):

- 48h/36h/24h Event Rule: Alert on Day d-2 OR Day d-1 -> 372/372 = 100.0% event detection.
- 12h Event Rule: Alert on Day d-1 specifically -> 355/372 = 95.4% event detection.
- Day-Level Recall: 93.41% (695 of 744 individual impending flare days flagged in test set).
- Clinical Workload: 0.77 alert days/pt/month (919 total alert days / 1200 patient-months in test set), representing individual alert days, not grouped multi-day episodes (~0.25-0.30 episodes/pt/mo).

Wearable Data Random Dropout Robustness:

- 0% Missing: ROC-AUC 0.9982 | Recall 93.4% | Precision 75.6% | F1 0.8358
- 10% Missing: ROC-AUC 0.9963 | Recall 78.8% | Precision 75.3% | F1 0.7700
- 20% Missing: ROC-AUC 0.9925 | Recall 61.0% | Precision 78.7% | F1 0.6874
- 30% Missing: ROC-AUC 0.9861 | Recall 37.9% | Precision 78.5% | F1 0.5113

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Transparent SHAP Attribution & Sensitivity Modeling

Slide 7: Explainability & What-If Sensitivity Simulator

- SHAP TreeExplainer Attribution: Local log-odds converted to probability percentage points.
  Mathematically reconciled: Cohort Reference Baseline (3.6%) + Bars = Displayed Risk%.
- 9 Clinical Factors: SpO2 vs baseline, resting HR, respiratory rate, sleep, steps, CAMS NAQI, dust, weather, adherence.
- Counterfactual What-If Simulator:
  - Allows interactive adjustment of indoor AQI reduction (0 to -150), adherence boost (+0 to +50%),
    and thermal swing dampening (0 to -8°C).
  - Shows model sensitivity on synthetic data; does NOT claim a proven clinical intervention effect.

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Editorial-Clinical Hospital Monitoring UI

Slide 8: Interactive Clinical Dashboard

- Editorial Design: Warm off-white theme, deep teal accent, serif headings, light/glass cards.
- 3D Lung Digital Twin: Three.js breathing model with respiratory rate pacing and risk tinting.
- 5 Vital Tiles: SpO2, HR, RR, Sleep, Steps with 7-day sparklines and baseline deviation delta.
- 365-Day Live Simulation Engine: Play/Pause/Speed controls, date scrubber, and 'Jump to Next Alert'.
- Light/Glass Longitudinal Telemetry: Deep teal SpO2, crimson HR, amber CAMS NAQI, and steps bars.
- Delhi GIS Station Map: 5 ambient stations with patient pinned and micro-climate readings.
- Error Boundary: Fault-tolerant architecture preventing blank screen crashes.

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Transparent Statement of Limitations

Slide 9: Honest Scientific & Clinical Limitations

1. Synthetic Cohort: Uses GOLD/GINA stage categories; proportions are our assumptions.
   Flare labels were assigned by our own drift simulation rules; metrics reflect rule recovery.
2. Synthetic Flare Rate: Averages ~3.64 flares/patient/year (1,821 events across 500 patients over 365 days;
   372 flares across 100 test patients), which is higher than typical real-world outpatient rates.
3. Uncapped Dust Hazard: In calculate_daily_flare_hazard, dust excess was not capped prior to multiplier
   calculation, though the composite daily hazard was bounded by an overall ceiling of 0.12 (12%/day).
4. No Real Patient Evidence: No prospective or retrospective evidence demonstrating safety or efficacy.
5. Modelled Weather & AQI: From Open-Meteo CAMS and ERA5 reanalysis models, not hardware monitoring stations.
6. Decision Support Only: Research prototype for decision-support; not a diagnostic tool.

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Project Deliverables & Repository Information

Slide 10: Assets, Credits & Public Access

- Author: Monalika Doda, SRM University Delhi-NCR, <MY TEAM NAME>, solo.
- Repository: https://github.com/MonalikaDoda/PulmoTwin
- Assets & Credits:
  - 3D Lung Model: 'Lungs' by prasanth.k on Sketchfab (CC BY 4.0, https://skfb.ly/oqOP7)
  - Weather & Air Quality: Open-Meteo.com (Copernicus CAMS & ECMWF ERA5) under CC BY 4.0
- Deliverables in docs/:
  - docs/PulmoTwin_Architecture_Diagram.pdf (Vector pipeline)
  - docs/PulmoTwin_Presentation_Deck.pdf (Executive pitch deck)
- License: MIT License (Copyright (c) 2026 Monalika Doda)
- Assistance Disclosure: Built with AI coding assistance (Antigravity).
- Quick-Start: Run backend on port 8000, frontend on port 3000. Step 3 is optional.